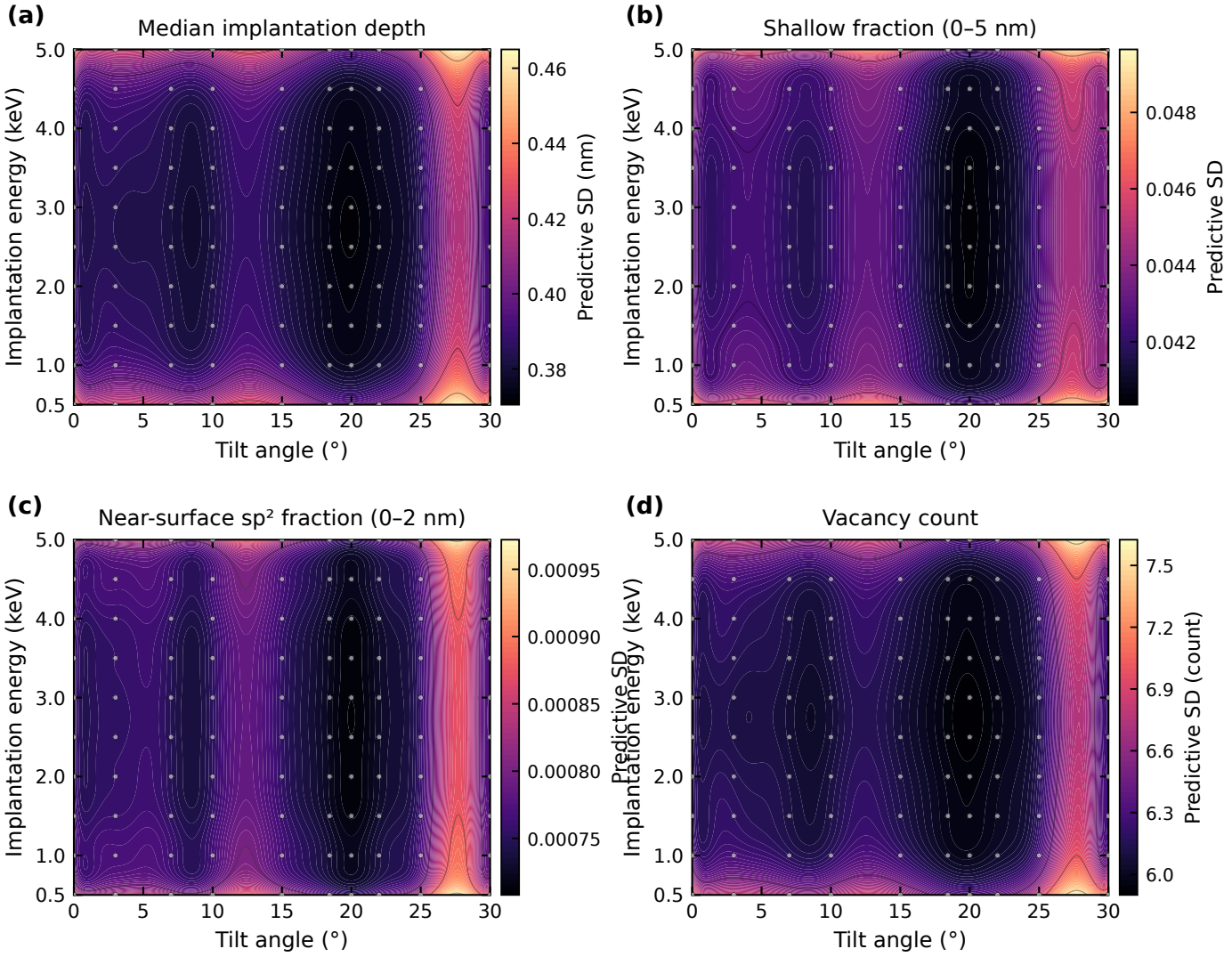

(a)
Median implantation depth
Implantation energy (keV)
Tilt angle (°)
Predictive SD (nm)
0.46
0.44
0.42
0.40
0.38
(b)
Shallow fraction (0–5 nm)
Implantation energy (keV)
Tilt angle (°)
Predictive SD
0.048
0.046
0.044
0.042
(c)
Near-surface sp² fraction (0–2 nm)
Implantation energy (keV)
Tilt angle (°)
0.00095
0.00090
0.00085
0.00080
0.00075
(d)
Vacancy count
Predictive SD
Implantation energy (keV)
Tilt angle (°)
Predictive SD (count)
7.5
7.2
6.9
6.6
6.3
6.0